

SIDHANT BANSAL

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing and building scalable distributed systems, full-stack applications, and data-intensive pipelines. Currently owning end-to-end maritime data infrastructure at the State of Washington, processing 10M+ messages and delivering predictive safety models. Strong foundation in algorithms, system design, and cloud architecture, with practical experience applying AI-assisted development tools in production workflows.

WORK EXPERIENCE

Washington State Department of Ecology

Seattle, WA

Software Engineer

Aug 2023 – Present

- Architected and owned a scalable Python data pipeline to simulate vessel traffic and movement patterns, ingesting and processing 10M+ AIS messages with high reliability and low latency.
- Implemented algorithms in Python and SQL to construct vessel tracks and automatically detect escort events, surfacing 2,000+ maritime incidents that inform state safety policy.
- Built predictive models to forecast vessel groundings, collisions, and spill risks directly strengthening Washington State maritime safety operations.
- Leveraged prompt engineering to accelerate ASP.NET/C# development across SPIIS, a state platform for environmental incident management, reducing implementation time by 30%.

Platform Science

San Diego, CA

Software Developer Intern

Jun 2022 – Sep 2022

- Engineered and shipped RESTful APIs using Node.js, Express, and Laravel (PHP) supporting backward compatibility and service versioning, improving client-side productivity by 20%.
- Refactored JavaScript/TypeScript and PHP codebases using design patterns and reusable components. Introduced automated unit tests via Docker-based CI, increasing development velocity by 15%.

Accenture

Bangalore, India

Application Developer

Sep 2019 – Aug 2021

- Built and scaled React.js/MEAN stack web apps and Node.js/Express backends, reducing API response time by 40% and supporting 30% more concurrent users via caching and optimized database indexing.
- Improved user engagement by 25% through UI/UX and API optimizations, partnered with PMs and stakeholders to ship features on time within Agile sprints.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL

Frameworks & Cloud: React, Angular, Node.js, Express, Spring Boot, Laravel, ASP.NET, AWS, Azure, Docker

Databases & Tools: PostgreSQL, MySQL, MongoDB, SQL Server, Git, JIRA, Unity, ArcGIS, GitHub Copilot

KEY PROJECTS

VR Therapy Platform: Graduate Research (EYE Research Group)

Sep 2022 - Jun 2023

- Consolidated four separate VR therapy applications into a unified Unity/C# platform, improving rendering performance by 20% and reducing code redundancy by 35% through modular refactoring.

Dependify

Apr 2022 - Jun 2022

- Architected a distributed microservices architecture (Java, Spring Boot, PostgreSQL, AWS EC2) for centralized dependency management, handling 10,000+ daily requests with 40% faster response time.

EDUCATION

University of Washington

Seattle, WA

Master of Science in Computer Science

Sep 2021 - Jun 2023

Coursework: Parallel Computing, Distributed Computing, Algorithms, Computer Graphics, Software Management

Guru Gobind Singh Indraprastha University

New Delhi, India

Bachelor of Technology in Information Technology

Aug 2015 - Jun 2019

Coursework: Data Structures and Algorithms, Operating System, Computer Graphics & Architecture, Cloud Computing